Exercício Autômatos - 27/10

Integrantes:

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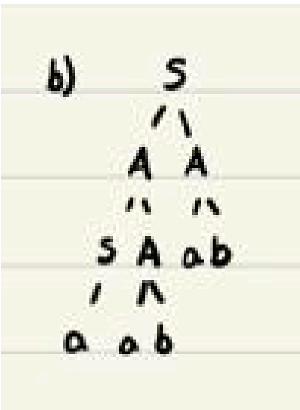
1)

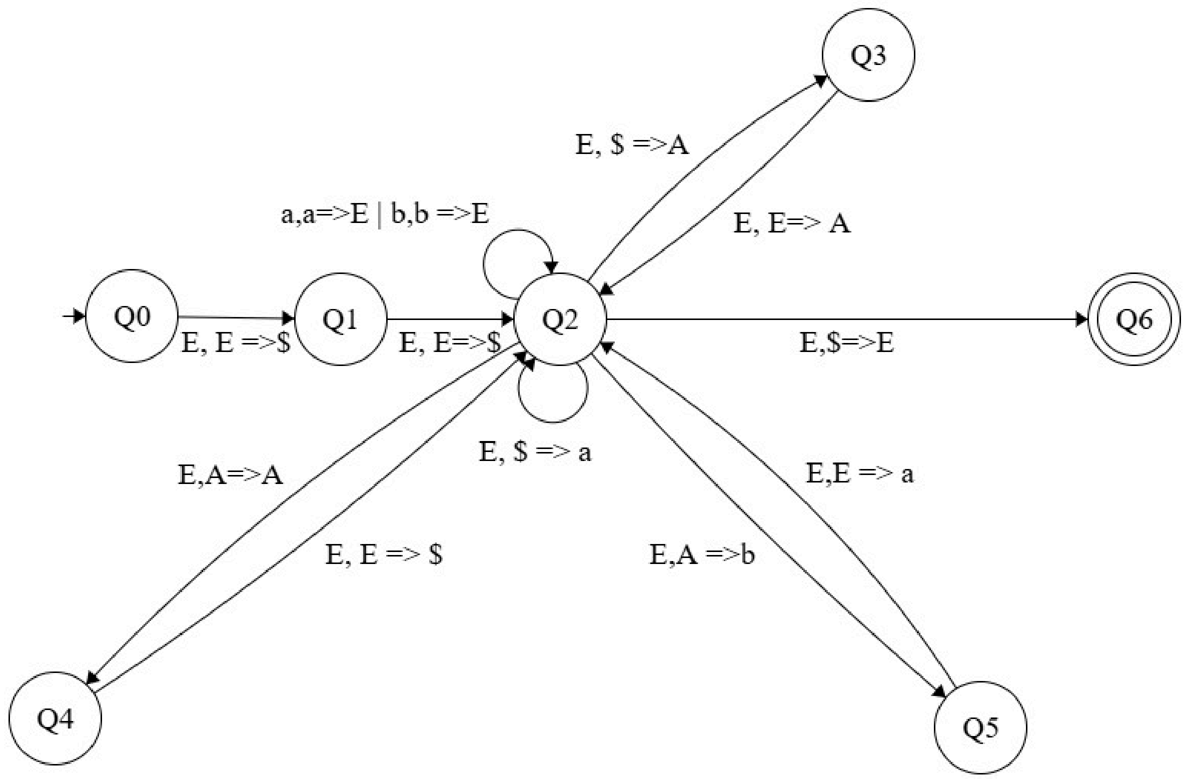
a)
Regras da Gramática:
S -> AA | a
A -> SA
A -> ab

Portanto: S => AA => SAab => aabab

b)



c)



d)

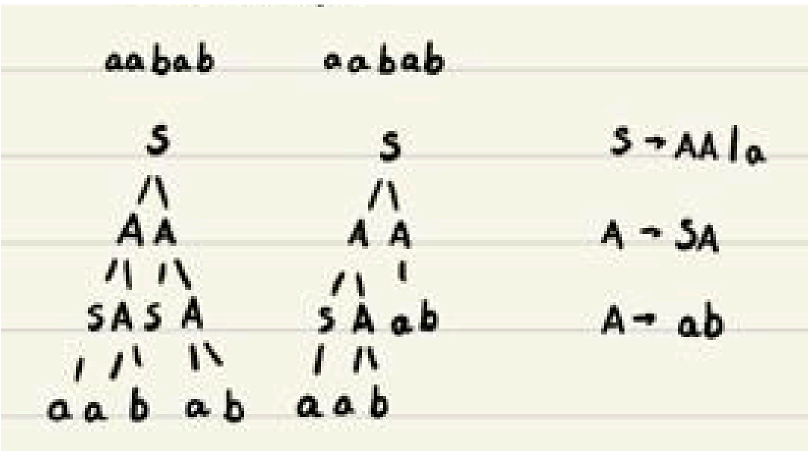
Cadeia ‘aabab’: ACEITA pelo autômato

(q0, aabab, ε)
|- (q1, aabab, $)
|- (q2, aabab, S$)
|- (q3, aabab, A$)
|- (q2, aabab, AA$)
|- (q4, aabab, AA$)
|- (q2, aabab, SAA$)
|- (q2, aabab, aAA$)
|- (q2, abab, AA$)
|- (q5, abab, bA$)
|- (q2, abab, abA$)
|- (q2, bab, bA$)
|- (q2, ab, A$)
|- (q5, ab, b$)
|- (q2, ab, ab$)
|- (q2, b, b$)
|- (q2, ε, $)
|- (q6, ε, ε)

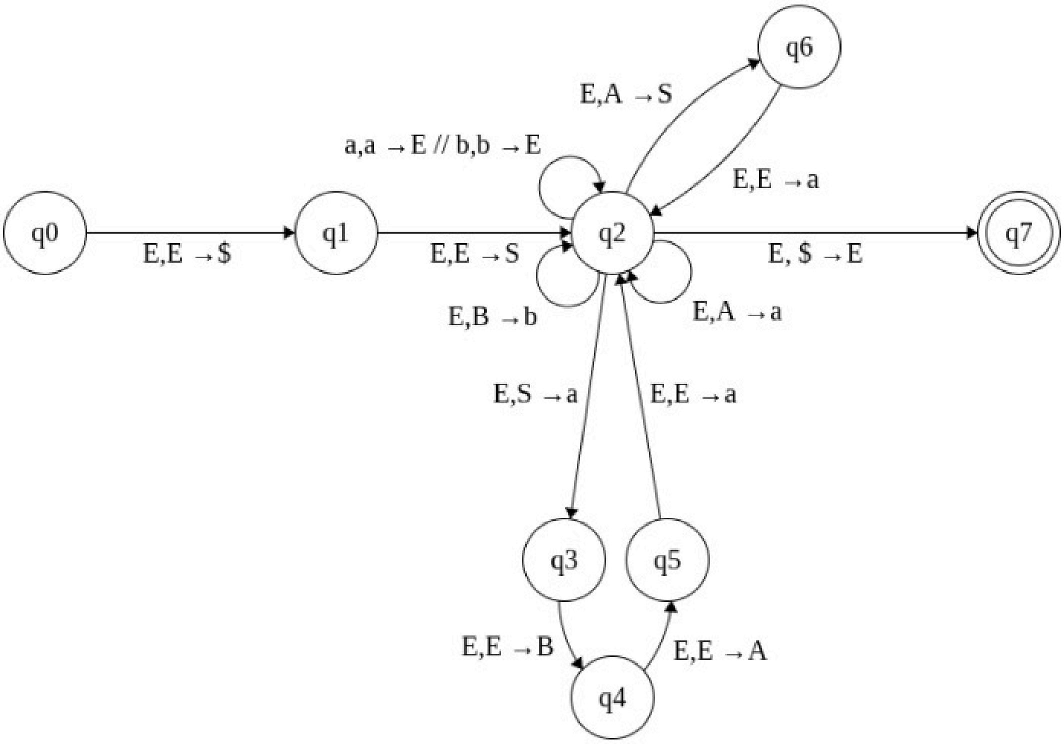
e)

Ambígua, 2 ou mais árvores para a mesma cadeia.

A cadeia ‘aabab’ é ambígua:



2)

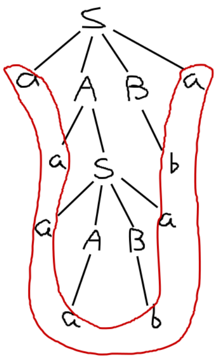


Cadeia ‘aaaababa’: ACEITA pelo autômato

(q0, aaaababa, ε)
|- (q1, aaaababa, $)

|- (q2, aaaababa, S$)
|- (q3, aaaababa, a$)
|- (q4, aaaababa, Ba$)
|- (q5, aaaababa, ABa$)
|- (q2, aaaababa, aABa$)
|- (q2, aaababa, ABa$)
|- (q2, aaababa, SBa$)
|- (q2, aaababa, aSBa$)
|- (q2, aababa, SBa$)
|- (q3, aababa, aSBa$)
|- (q4, aababa, BaBa$)
|- (q5, aababa, ABaBa$)
|- (q2, aababa, aABaBa$)
|- (q2, ababa, ABaBa$)
|- (q2, ababa, aBaBa$)
|- (q2, baba, BaBa$)
|- (q2, baba, baBa$)
|- (q2, aba, aBa$)
|- (q2, ba, Ba$)
|- (q2, ba, ba$)
|- (q2, a, a$)
|- (q2, ε, $)
|- (q7, ε, ε)

Árvore Sintática:



3)
Q0 = A
Q1= B
Q2 = C
Q3 = D
Q4 = E

A -> 0B
A -> 1D
B -> 0C
B -> 0

B -> 1C
B -> 1
C -> 0B
C -> 1B
D -> 0E
D -> 0
D -> 1E
D -> 1
E -> 0D
E -> 1D

4)